As an alternative, the group of points from the input signal can be chosen *symmetrically* around the output point:

$$y[80] = \frac{x[78] + x[79] + x[80] + x[81] + x[82]}{5}$$

This corresponds to changing the summation in Eq. 15-1 from: $j = 0$ to $M-1$, to: $j = -(M-1)/2$ to $(M-1)/2$. For instance, in an 11 point moving average filter, the index, j, can run from 0 to 11 (one side averaging) or -5 to 5 (symmetrical averaging). Symmetrical averaging requires that M be an *odd* number. Programming is slightly easier with the points on only one side; however, this produces a relative shift between the input and output signals.

You should recognize that the moving average filter is a *convolution* using a very simple filter kernel. For example, a 5 point filter has the filter kernel: $\cdots 0, 0, 1/5, 1/5, 1/5, 1/5, 1/5, 0, 0 \cdots$. That is, the moving average filter is a convolution of the input signal with a *rectangular pulse* having an area of *one*. Table 15-1 shows a program to implement the moving average filter.

```
100 'MOVING AVERAGE FILTER
110 'This program filters 5000 samples with a 101 point moving
120 'average filter, resulting in 4900 samples of filtered data.
130 '
140 DIM X[4999]                    'X[ ] holds the input signal
150 DIM Y[4999]                    'Y[ ] holds the output signal
160 '
170 GOSUB XXXX                     'Mythical subroutine to load X[ ]
180 '
190 FOR I% = 50 TO 4949            'Loop for each point in the output signal
200   Y[I%] = 0                    'Zero, so it can be used as an accumulator
210   FOR J% = -50 TO 50           'Calculate the summation
220     Y[I%] = Y[I%] + X(I%+J%]
230   NEXT J%
240   Y[I%] = Y[I%]/101            'Complete the average by dividing
250 NEXT I%
260 '
270 END
```

TABLE 15-1

Noise Reduction vs. Step Response

Many scientists and engineers feel guilty about using the moving average filter. Because it is so very simple, the moving average filter is often the first thing tried when faced with a problem. Even if the problem is completely solved, there is still the feeling that something more should be done. This situation is truly ironic. Not only is the moving average filter very good for many applications, it is *optimal* for a common problem, reducing random white noise while keeping the sharpest step response.